Convex optimization for machine learning – TD1

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1 Exercises

Exercise 1. (Banach-Picard's fixed point theorem)

Let E be a complete metric space and $f : E \to E$ a **contracting** mapping, that is:

$$\exists \rho \in [0, 1[,\ \forall x, y \in E,\ \mathrm{d}\big(f(x), f(y)\big) \leq \rho \mathrm{d}(x, y).$$

Prove that f admits **one unique** fixed point[1] x^*, and that the sequence defined by:

$$\begin{cases} x_0 \in E \\ x_{n+1} = f(x_n),\ \forall n \in \mathbb{N}. \end{cases}$$

converges to x^* with rate of convergence ρ^n, *i.e.*:

$$\forall n \in \mathbb{N},\ \mathrm{d}(x_n, x^*) \leq \rho^n \mathrm{d}(x_0, x^*).$$

Exercise 2. (Gradient calculus)

1. Calculate the gradient of the following functions. A, M and Q are fixed matrices, and $\mathbf{b}$ is a fixed vector. More precisely, M belongs to $\mathcal{M}_{m,n}(\mathbb{R})$ and Q to $\mathcal{M}_{p,n}(\mathbb{R})$.

   (a)
   $$\begin{array}{rccc} f_1 & : & \mathbb{R}^n & \to & \mathbb{R} \\ & & \mathbf{x} & \mapsto & \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2 \end{array}$$

   (b)
   $$\begin{array}{rccc} f_2 & : & \mathbb{R}^n & \to & \mathbb{R} \\ & & \mathbf{x} & \mapsto & \sum_{i=1}^n \log(1 + \exp(x_i)) \end{array}$$

   (c)
   $$\begin{array}{rccc} f_3 & : & \mathcal{M}_{m,p}(\mathbb{R}) & \to & \mathbb{R} \\ & & P & \mapsto & \frac{1}{2}\|M - PQ\|_F^2 \end{array}$$

2. Let g_1, g_2 and g_3 be functions such that $g_1 : \mathbb{R}^{n_1} \to \mathbb{R}^{n_2}$, $g_2 : \mathbb{R}^{n_2} \to \mathbb{R}^{n_3}$ and $g_3 : \mathbb{R}^{n_3} \to \mathbb{R}$. Let:

   $$f_4 = g_3 \circ g_2 \circ g_1.$$

   Compute the gradient of f_4 using the Jacobian matrices of g_i for $i \in [\![1, 3]\!]$.

[1] A point such that $f(x^*) = x^*$.

3. Suppose that computing one element of the Jacobian matrices costs C_J and that multiplying two numbers costs C_M. How much does it cost to compute $\nabla f_4(x)$?

Exercise 3. Let f be a $\mathcal{C}^2$ function from $\mathbb{R}^d$ to $\mathbb{R}$. Assume there exist $\mu, L \in \mathbb{R}_+$ such that[2]

$$\mu I_d \preceq \nabla^2 f(\mathbf{x}) \preceq L I_d.$$

1. Let $M \in \mathcal{S}(\mathbb{R}^d)$ be a real symmetric matrix satisfying the previous assumption and note $\|\cdot\|_2$ its operator matrix with respect to the Euclidean norm on $\mathbb{R}^d$.

   (a) Let λ_1 be the greatest eigenvalue of M. Show that

   $$\|M\|_2 = |\lambda_1|.$$

   (b) Prove that:

   $$\mu \leq \|M\|_2 \leq L.$$

2. Show that the fixed point operator $F : \mathbf{x} \mapsto \mathbf{x} - \gamma \nabla f(\mathbf{x})$ is contractant for every γ in $]0, 2/L[$.

3. Show that the gradient method converges linearly[3] to some x^*.

4. How many iterations are necessary to ensure that $\|x_k - x^*\| \leq \varepsilon$?

[2] Where $\preceq$ denotes the partial order known as *Loewner order*: if M, N are positive semi-definite matrices, $M \preceq N$ if $N - M$ is positive semi-definite.

[3] That is:

$$\exists \rho \in]0,1[, \ \lim_{n \to \infty} \big(\mathrm{d}(x_{n+1}, x^*)\big)^{\frac{1}{n}} \leq \rho.$$